Efficient Layout-Aware Document Understanding for Educational Transcripts via Systematic Model Compression

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Abstract

Deploying layout-aware transformer models for educational document processing presents a fundamental tension between accuracy requirements and computational constraints. We investigate model compression techniques for educational transcript information extraction through systematic evaluation of knowledge distillation, pruning, and quantization applied to state-of-the-art layout-aware architectures. Our experiments reveal that compression success critically depends on initialization strategy: pre-trained student models maintain or even exceed teacher performance with substantial parameter reduction, while randomly initialized students exhibit complete training failure. We identify structured attention head pruning as uniquely beneficial, while feed-forward network pruning proves detrimental. Dynamic quantization provides superior accuracy-memory trade-offs compared to static approaches. These findings establish that effective compression of layout-aware models requires careful consideration of pre-training dependencies and architectural component roles, providing practical insights for institutions seeking to balance accuracy with computational efficiency.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Machine learning*.

Keywords

Document understanding, Model compression, Knowledge distillation, Educational technology, Layout-aware models

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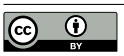
1 Introduction

Educational institutions process large volumes of academic transcripts for verification, credit transfer, and compliance [17]. The structural complexity of these documents—combining tabular grade data with diverse institutional formatting conventions and critical spatial relationships—presents unique challenges for automated processing. Traditional Optical Character Recognition (OCR) pipelines that convert documents to plain text fail to preserve layout information essential for accurate understanding [7, 22]. Layout-aware transformers, particularly the LayoutLM family [6, 22, 23], address this challenge by jointly modeling textual and visual features. However, models such as LayoutLMv3 contain approximately 133 million parameters [6], limiting deployment in resource-constrained settings. Model compression techniques—knowledge distillation [5], pruning [3], and quantization [14]—offer potential solutions, yet their effects on multimodal document models remain underexplored. We investigate how these techniques impact layout-aware models for transcript information extraction.

This study investigates: *How do compression techniques affect layout-aware models for educational transcript processing, and what determines successful compression?* First, We compare LayoutLMv2, LayoutLMv3, and LiLT for transcript extraction, establishing baselines in accuracy and efficiency. Second, we evaluate knowledge distillation strategies, showing that pre-trained student initialization is essential: it preserves or even surpasses teacher performance with significant size reduction, while random initialization fails. Third, we assess structured and unstructured pruning as well as static and dynamic quantization and show that structured attention head pruning improves both accuracy and throughput, whereas MLP pruning degrades performance. Finally, we clarify trade-offs among the compression methods and their impact on downstream transcript processing.

2 Related Work

Layout-aware document understanding has advanced from rule-based systems to multimodal transformers that jointly model text and visual structure. The LayoutLM family [6, 22, 23] integrates 2D positional embeddings and visual features, with LayoutLMv3 [6] achieving strong results through unified text–image masking. Other models, including LiLT [21] and FormNet [10] address multilingual transfer and structural reasoning. However, these approaches have not been systematically studied for educational transcripts, and their computational cost limits practical deployment. *Model compression techniques* such as knowledge distillation [5], pruning [3, 11], and quantization [8, 14] improve efficiency. DistilBERT [18] and TinyBERT [9] demonstrate effective teacher-initialized distillation, while intermediate alignment [16], attention transfer [24], and multi-step methods [13] further enhance performance. Yet, their impact on multimodal layout-aware models remains underexplored. *Finally, Educational documents* contain heterogeneous



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Table 1: Dataset composition

Split	Docs	Tokens
Training	126	24017
Validation	42	8934
Test	43	8311
Total	211	41262

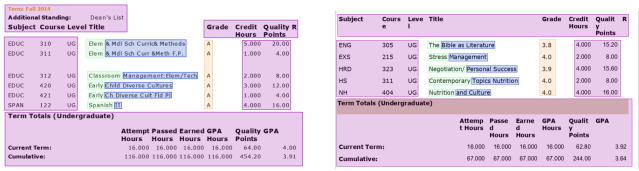
layouts and dense structured data, demanding high extraction accuracy. Existing solutions are often institution-specific [4], and prior educational research has focused more on assessment and analytics [1, 17] than structural document understanding. We address this gap by evaluating compression strategies for transcript processing while preserving required accuracy.

3 Methodology

Data Collection: We constructed a dataset of 211 academic transcript page images representing diverse formatting conventions across universities from different regions and educational systems. Each sample corresponds to a single page, with multi-page transcripts split so that each page is treated as an independent document, reflecting common document processing workflows. The dataset captures variation in header structures, tabular layouts, grading schemes, and overall visual organization to support robust evaluation across transcript styles. The annotated dataset was partitioned using stratified sampling with a 60/20/20 split for training, validation, and testing; Table 1 summarizes the distribution. *All documents were manually anonymized* to remove personally identifiable information (names, IDs, addresses) and institutional identifiers, and the dataset will be released for research use subject to appropriate data sharing agreements.

Annotation Protocol and Quality Control: Token-level annotations were created using Label Studio [20], focusing on three entity categories: course identifiers and titles (B-COURSE/I-COURSE), individual course grades (GPA, including letter grades and numerical scores on the 0.0–4.0 scale), and background tokens (O), which include headers, formatting elements, and administrative text. Figure 1 shows two representative transcript pages illustrating the diversity of institutional layouts and grading formats in the dataset. All 211 transcript pages were annotated by a single trained annotator to maintain consistent labeling decisions. To ensure quality, a second researcher periodically reviewed annotated documents, verified adherence to the guidelines, and provided feedback on ambiguous cases. This two-stage process helped resolve edge cases while maintaining labeling consistency across the dataset.

Data Preprocessing and Formatting: Each transcript page was converted into a structured format for layout-aware transformers. Text and bounding boxes were extracted using Tesseract OCR [19], producing hOCR files that preserve layout information. These were converted to line-delimited JSON containing the image path, OCR tokens, normalized bounding boxes $[x_{\min}, y_{\min}, x_{\max}, y_{\max}] \in [0, 1]^4$, and BIO labels. Pages exceeding the 512-token limit were processed with a sliding window (stride 128). Tokens in overlapping regions were assigned label -100 to ensure each token is evaluated once. Multi-page transcripts were treated as independent samples, reflecting page-level processing pipelines.



(a) Letter-grade transcript (b) Numerical-grade transcript

Figure 1: Zoomed annotated transcript regions showing COURSE and GPA token labels. Green/blue highlights indicate course-related tokens, orange highlights indicate grade or GPA-related tokens, and purple highlights indicate other tokens.

4 Experimental Design and Methods

Experimental Configuration: Experiments were conducted on a workstation with an NVIDIA RTX 6000 Ada GPU (48GB) and an AMD Ryzen Threadripper PRO 5975WX CPU running Ubuntu 22.04.4 with CUDA 12.4. Models were implemented using PyTorch 2.6.0 and ONNX Runtime 1.22.1. GPU benchmarks were used for training and inference, while CPU benchmarks were performed only for quantization evaluation. All models used FP16 mixed-precision training and identical preprocessing pipelines to ensure fair comparison.

Evaluation Methodology: We evaluate sequence labeling performance using seqeval [15], reporting micro-averaged precision, recall, F1-score, and token-level accuracy. F1-score is the primary metric as it balances precision and recall for transcript extraction tasks. Padding and overlapping tokens are masked with label value -100 to ensure each token is evaluated once. Throughput (samples/s) measures end-to-end page processing latency, as shown in Equation (1), where N_{pages} denotes the number of processed test pages and T_{total} represents the total wall-clock processing time.

$$\text{Throughput} = \frac{N_{\text{pages}}}{T_{\text{total}}} \tag{1}$$

Knowledge Distillation Framework: Student models follow the LayoutLMv3 architecture with reduced encoder depth (6, 8,10 layers), a common compression strategy [9]. Key parameters— embedding size (768), attention heads (12), and feed-forward dimensions (3072)—remain unchanged to preserve representational capacity. Models were trained for 50 epochs with early stopping based on validation F1-score (p = 10). We compare two initialization strategies: using pretrained LayoutLMv3 weights (retaining early-layer representations after layer truncation) and random initialization. Each strategy is evaluated with and without knowledge distillation to isolate the impact of teacher guidance.

The overall training objective for knowledge distillation (KD) combines three complementary loss components to balance task performance, knowledge transfer, and representational alignment between teacher and student models. The total loss is defined in Equation 2 where $\alpha = 0.6$ balances task and distillation losses, $\beta = 0.05$ weights the hint loss.

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{task}} + (1 - \alpha)\mathcal{L}_{\text{KD}} + \beta \mathcal{L}_{\text{hint}} \tag{2}$$

$$\mathcal{L}_{\text{task}} = -\frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} w_{y_i} \left[(1-\epsilon) \log p_s^{(i)}(y_i) + \frac{\epsilon}{C} \sum_{c=1}^{C} \log p_s^{(i)}(c) \right] \tag{3}$$

$$\mathcal{L}_{\text{KD}} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \sum_{c=1}^{C} p_t^{(i)}(c) \log \frac{p_t^{(i)}(c)}{p_s^{(i)}(c)} \cdot T^2 \tag{4}$$

$$\mathcal{L}_{\text{hint}} = \frac{1}{|\mathcal{P}|} \sum_{(j,k) \in \mathcal{P}} \frac{1}{|\mathcal{V}| \cdot d_h} \sum_{i \in \mathcal{V}} \| h_s^{(k,i)} - W h_t^{(j,i)} \|_2^2 \tag{5}$$

Equation 3,4 and 5 denotes the individual losses where $\mathcal{V} = \{i \mid \text{mask}_i = 1 \wedge y_i \neq -100\}$ represents valid token positions, $\mathcal{P}$ denotes layer pairs determined through proportional mapping, w_{y_i} are class weights, $\epsilon = 0.1$ is the label smoothing parameter (applied only during training), $T = 3.0$ is the temperature, C is the number of classes, d_h is the hidden dimension, $p_s^{(i)}(c) = \text{softmax}(z_s^{(i)}/T)_c$ and $p_t^{(i)}(c) = \text{softmax}(z_t^{(i)}/T)_c$ are temperature-scaled probability distributions, $h_s^{(k,i)}$ and $h_t^{(j,i)}$ are hidden states from student layer k and teacher layer j at position i, and $W \in \mathbb{R}^{d_h \times d_h}$ is a learned linear projection initialized as identity. The task loss uses cross-entropy loss with class-specific weights w_{y_i} addressing class imbalance, where entity classes (B-COURSE, I-COURSE, GPA) receive weight 1.5 and background class (O) receives weight 1.0. Label smoothing with parameter $\epsilon = 0.1$ is applied during training only to prevent overconfident predictions. Only valid tokens contribute to loss computation. The knowledge distillation loss applies temperature-scaled Kullback-Leibler divergence between teacher and student probability distributions, computed only over valid token positions. The KL divergence is computed per-token then averaged over $\mathcal{V}$. The T^2 scaling factor maintains gradient magnitudes across different temperature values. The hint loss computes mean squared error between intermediate hidden states of matched teacher-student layer pairs [16]. For multimodal inputs containing text tokens and visual patch embeddings, only text token representations (first S_{text} positions where S_{text} is the input sequence length) are used for hint loss computation. The projection matrix W aligns dimensionalities and is optimized jointly with student parameters.

Student layers are paired with teacher layers using proportional mapping. For a student with N_S layers and teacher with N_T layers, student layer s maps to teacher layer $t = \text{round}(s \cdot N_T/N_S)$, while embedding layers are paired directly. All losses respect attention masks and ignore indices (-100), ensuring padded tokens do not affect gradients. Teacher parameters remain frozen during training. Each configuration was trained with five random seeds for both pretrained and randomly initialized students to ensure robustness.

Class Imbalance Mitigation: The dataset is highly imbalanced: background tokens (O) account for 84% of tokens, while COURSE entities represent 13% and GPA entities 3%. To prevent the model from favoring the majority class, we apply class-weighted cross-entropy loss [2], assigning weight 1.5 to entity classes (B-COURSE, I-COURSE, GPA) and 1.0 to background tokens. Fixed weights provide stable training because transcript pages vary widely in token distribution (e.g., dense course tables vs. mostly background pages). They also prioritize accurate extraction of courses and grades, which are critical for transcript verification, over balanced performance across all token types.

Post-Distillation Compression: After knowledge distillation, we apply additional compression techniques to the optimal pre-trained student models to improve inference efficiency and memory usage. We evaluate both structured and unstructured pruning. Unstructured pruning removes low-magnitude weights using global L1-norm scoring across linear and convolutional layers (excluding embeddings and classifiers) with sparsity levels of 30% and 50%. We apply one-shot pruning without retraining to measure the immediate impact of weight removal [3]. Structured pruning removes entire architectural components, including attention heads and MLP neurons. Specifically, 25% of attention heads are removed using L2-norm scoring across head dimensions, and 40% of MLP neurons are pruned along the intermediate layer dimension. For further compression, we evaluate post-training INT8 quantization using ONNX Runtime 1.21.1. Static quantization determines fixed parameters through calibration on 126 training samples using Min-Max calibration, while dynamic quantization computes activation scales during inference. Both methods quantize MatMul and Gemm operations with per-channel weight quantization.

5 Experimental Results and Analysis

We evaluate model compression for transcript processing through four key contributions outlined in Section 1, guiding analysis from baseline comparison to deployment considerations.

5.1 Baseline Model Performance Comparison

To establish benchmarks and identify an optimal teacher model, we evaluated three state-of-the-art layout-aware document understanding models on our transcript dataset. Table 2 reports F1-score, model size, and throughput for the base models. LayoutLMv3 provides the best accuracy–efficiency trade-off, achieving the highest F1-score (0.9507) with a smaller footprint than LayoutLMv2 (478.10MB vs. 764.82MB). Although LiLT shows higher throughput, its lower F1-score (0.9049) makes it less suitable for high-accuracy transcript extraction. Therefore, we select LayoutLMv3 as the teacher model for subsequent distillation experiments.

Table 2: Baseline model comparison

Model	F1	MB	S/s
LayoutLMv3	**0.95**	**478.1**	44.66
LayoutLMv2	0.91	764.8	46.98
LiLT	0.90	496.5	**92.58**

5.2 Knowledge Distillation Analysis

We evaluated knowledge distillation using a fine-tuned 12-layer LayoutLMv3 as teacher. For student architectures of 6, 8, and 10 layers, we tested four training configurations: with/without pre-trained initialization and with/without distillation. This design isolates the contributions of pre-trained weights versus knowledge transfer. Table 3 shows results across all configurations. The key finding is that pre-trained initialization is essential for successful

knowledge distillation in layout-aware document understanding. Models with random initialization perform poorly ($F1 \approx 0.31$) even with teacher supervision, indicating that our 211-sample dataset is insufficient to learn cross-modal representations from scratch. This suggests that effective distillation requires pretrained visual–textual representations rather than learning them solely from limited task data.

Table 3: Impact of student initialization

L	Pre.	Dist.	F1	MB
6	N	Y	0.3254 ± 0.0118	315.87
6	N	N	0.3187 ± 0.0180	315.87
6	Y	Y	0.9355 ± 0.0072	315.87
6	Y	N	0.9313 ± 0.0096	315.87
8	N	Y	0.3436 ± 0.0214	369.95
8	N	N	0.3355 ± 0.0233	369.95
8	**Y**	**Y**	**0.9475 ± 0.0023**	**369.95**
8	Y	N	0.9499 ± 0.0034	369.95
10	N	Y	0.3398 ± 0.0200	424.02
10	N	N	0.3387 ± 0.0245	424.02
10	**Y**	**Y**	**0.9540 ± 0.0052**	**424.02**
10	Y	N	0.9458 ± 0.0110	424.02

Notes: L = Layers; Pre. = Pre-trained; Dist. = Distillation; Y/N = Yes/No. Teacher: F1 = 0.95, Size = 478.1 MB. Results are mean ± std over 5 seeds.

Among pre-trained students, the results show effective compression with minimal performance loss. The 10-layer distilled model achieves an F1-score of 0.9540, slightly surpassing the teacher's 0.9507 while reducing model size by 11.3% (478.10 MB to 424.02 MB). The 8-layer distilled model achieves greater compression at 22.6% (369.95 MB) while maintaining an F1-score of 0.9475. The small difference between these configurations (0.65 percentage points) suggests comparable effectiveness despite different compression levels.

Direct fine-tuning without distillation yields performance comparable to distillation. The 8-layer model achieves an F1-score of 0.9499, while the 10-layer model reaches 0.9458. The small differences between distilled and non-distilled models (0.24–0.82 points) suggest that most gains come from pre-trained initialization rather than soft-label distillation. This indicates that substantial compression can be achieved through architectural reduction alone, with the choice between 8- and 10-layer models depending on the desired balance between compression and accuracy.

To analyze how compression affects specific extraction tasks, we report entity-level performance for COURSE and GPA entities in Table 4. The results show asymmetric degradation under random initialization: COURSE extraction F1-score drops to 0.40, while GPA extraction F1-score stays around 0.62. This difference arises because GPA values are short and consistently formatted (e.g., "A", "B"), making them easier to learn from limited data, whereas COURSE entities vary in length and formatting across transcripts and therefore require stronger pretrained representations.

Among pre-trained models, both compressed configurations maintain strong performance on both entity types. The 8-layer distilled model achieves a COURSE F1-score of 0.9853 and perfect GPA extraction (1.0000), while the 10-layer model achieves

Table 4: Entity-wise performance

L	Pre.	Dist.	Entity F1	
			COURSE	GPA
6	N	Y	0.3925 ± 0.0163	0.6211 ± 0.0171
6	N	N	0.3691 ± 0.0262	0.6288 ± 0.0302
6	Y	Y	0.9772 ± 0.0026	0.9903 ± 0.0064
6	Y	N	0.9783 ± 0.0055	0.9905 ± 0.0074
8	N	Y	0.4099 ± 0.0142	0.6232 ± 0.0351
8	N	N	0.3899 ± 0.0220	0.6307 ± 0.0303
8	**Y**	**Y**	**0.9853 ± 0.0065**	**1.0000 ± 0.0000**
8	Y	N	0.9822 ± 0.0047	1.0000 ± 0.0000
10	N	Y	0.4011 ± 0.0223	0.6308 ± 0.0251
10	N	N	0.4010 ± 0.0253	0.6355 ± 0.0306
10	**Y**	**Y**	**0.9845 ± 0.0045**	**0.9981 ± 0.0043**
10	Y	N	0.9788 ± 0.0045	1.0000 ± 0.0000

COURSE F1 of 0.9845 and GPA F1 of 0.9981. Both compressed models slightly outperform the teacher's COURSE F1 of 0.9837. Across all pre-trained models, GPA extraction consistently achieves higher F1-scores (0.99–1.00) than COURSE extraction (0.97–0.99), reflecting the greater difficulty of identifying course boundaries in diverse transcript layouts.

5.3 Post-Distillation Compression Analysis

After knowledge distillation, we evaluate additional compression techniques on the optimal pre-trained student models to improve inference efficiency and memory usage. Specifically, we compare unstructured magnitude-based pruning with structured pruning of attention heads and feed-forward networks. Table 5 summarizes their impacts.

The results show clear differences between pruning strategies. Structured attention head pruning is the most effective, with the 8-layer model removing 25% of heads while maintaining strong performance (F1 = 0.9293 vs. 0.9472 baseline) and achieving a 1.80× throughput improvement. This suggests redundancy in attention mechanisms for transcript extraction, where local spatial reasoning within table rows is more important than long-range dependencies [12]. In contrast, MLP pruning at 40% sparsity causes severe degradation, reducing F1-scores below 0.03. This indicates that feed-forward layers encode critical semantic mappings that cannot be compensated for by attention alone.

Table 5: Pruning Effectiveness Analysis

L	Pruning Strategy	F1	MB	S/s
6	Baseline	0.9368	315.87	43.066
6	Unstructured (30%)	0.8247	315.87	88.495
6	Unstructured (50%)	0.1025	315.87	88.823
6	Structured (Attn 25%)	0.9010	315.87	86.770
6	Structured (MLP 40%)	0.0195	315.87	88.063
8	Baseline	0.9472	369.95	40.433
8	Unstructured (30%)	0.8767	369.95	73.104
8	Unstructured (50%)	0.0687	369.95	76.399
8	**Struct. (Attn 25%)**	**0.9293**	**369.95**	**72.891**
8	Structured (MLP 40%)	0.0159	369.95	74.635
10	Baseline	0.9526	424.02	36.224
10	Unstructured (30%)	0.8847	424.02	61.576
10	Unstructured (50%)	0.1324	424.02	63.375
10	Structured (Attn 25%)	0.9156	424.02	60.913
10	Structured (MLP 40%)	0.0224	424.02	62.065

Unstructured magnitude pruning shows intermediate behavior. At 30% sparsity, models experience moderate performance drops (6–7 points for larger models) but gain 70–105% throughput. However, at 50% sparsity performance collapses (F1 < 0.15), suggesting a critical threshold where essential features are lost. Additionally, irregular sparsity patterns limit hardware acceleration on standard CPUs, making structured pruning more practical for deployment.

We evaluated INT8 quantization for CPU deployment, with results shown in Table 6. Dynamic INT8 quantization consistently outperforms static quantization, achieving 38–48% memory reduction while maintaining higher accuracy. The 8-layer dynamic model provides the best balance, retaining 99.9% of baseline performance (F1 = 0.9458) while reducing memory by 43.6% and improving throughput by 28.8%. Dynamic quantization performs better because it adapts activation scales during inference, while static quantization fixes scales during calibration. Since document images vary in layout and text density, fixed calibration scales can lead to accuracy loss. Static quantization in our setup was calibrated on only 126 samples using MinMax calibration, which is sensitive to outliers and may cause distribution mismatch.

Table 6: INT8 quantization CPU

L	Quant.	F1	MB	S/s
6	Baseline	0.9368	315.87	7.488
6	Static INT8	0.8978	194.89	8.832
6	Dynamic INT8	0.9098	194.87	10.796
8	Baseline	0.9472	369.95	6.569
8	Static INT8	0.9274	208.62	6.813
8	**Dynamic INT8**	**0.9458**	**208.60**	**8.462**
10	Baseline	0.9526	424.02	5.093
10	Static INT8	0.5445	222.36	5.843
10	Dynamic INT8	0.9385	222.33	7.291

Quantization sensitivity also varies by model depth. The 10-layer model shows severe degradation under static quantization (F1 = 0.5445), while the 6-layer model degrades more gradually (F1 = 0.8978), suggesting that quantization errors accumulate across deeper transformer layers.

6 Conclusion

This study provides systematic evaluation of compression techniques for layout-aware document understanding models applied to educational transcript processing. Our experiments on 211 annotated transcript pages reveal several key findings: Pre-trained initialization is essential for successful compression, maintaining teacher-level accuracy with substantial size reduction, while random initialization causes catastrophic failure. Structured attention head pruning can simultaneously improve both accuracy and throughput, whereas MLP pruning proves detrimental. Dynamic quantization outperforms static approaches, providing superior accuracy-memory trade-offs for variable document layouts. These compression techniques enable substantial model size reduction while maintaining high accuracy, though validation in production environments with larger datasets remains necessary before operational deployment.

Future work will evaluate these approaches on larger, multi-domain datasets to strengthen generalization claims, investigate progressive distillation and hybrid compression strategies that combine multiple techniques, and conduct deployment studies in educational institutions to assess real-world performance and integration challenges.

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